



CURSO: MATEMÁTICA II

COD. CURSO: MA-123MNOP

TIPO DE PRUEBA:

PRACTICA No.

1

Ex. PARCIAL

EX. FINAL

EX. SUST.

Calcule las integrales

$$\int \frac{1}{x} dx = \ln |x| + C$$

1. $\int \frac{(x^2 - 1)dx}{(x^2 + 1)\sqrt{x^4 + 1}}$

2. $\int \frac{\sqrt{x} \arcsen(\sqrt{x})}{\sqrt{x^2 - x^3}} dx$

3. $\int \frac{x e^{\arctan(x)}}{(1 + x^2)^{3/2}} dx$

4. $\int \frac{(x + x \tan^2(x))dx}{3 + 4 \tan(x) + \sec^2(x)}$

5. $\int \frac{(\tan(x) + \tan^3(x))dx}{(4 + \sec^2(x))^3 \cdot (\tan^2(x) + 8)^4}$

6. $\int \frac{(3e^{12x} - 2e^{6x})dx}{e^{12x} - 5e^{6x} + 6}$

7. $\int \frac{(\ln^3(x) + 2\ln(x) - 1)dx}{x(\ln^2(x) - 4\ln(x) + 8)^4}$

8. $\int \frac{\cos(x)dx}{3 - 3\sin(x) + 2\cos(x)}$

$$t^2 - 5t + 6$$

$$(t-2)(t-3)$$

$$e^{6x} = t$$

$$d(e^{6x}) = d(t)$$

Los profesores del curso

$$e^{6x} \cdot 6 = dt$$

$$6e^{6x} dx = dt$$

$$dx = \frac{dt}{6e^{6x}}$$

$$6e^{6x}$$

$$dx = \frac{dt}{6e^{6x}}$$



N° _____

UNIVERSIDAD NACIONAL DE INGENIERIA
Facultad de Ingeniería Eléctrica y ElectrónicaCICLO ACADÉMICO: 2017 - 2CURSO: MATEMÁTICA II COD. CURSO: MA-123 PSECCIÓN: P

TIPO DE PRUEBA:

PRACTICA N° ☒ EX. PARCIAL ☐ EX. FINAL ☐ EX. SUST. ☐

NOTA	NÚMEROS	LETRAS

21/04/2017

N° _____

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FIRMA DEL PROFESOR

1) CALCULAR

$$E = \int \frac{(x^2-1)dx}{(x^2+1)\sqrt{x^4+1}}$$

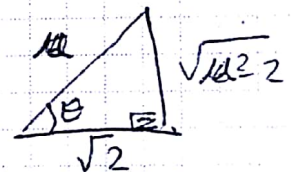
SOLUCION

TRANSFORMANDO EL INTEGRANDO

$$E = \int \frac{(x^2-1)dx}{x(x+\frac{1}{x})\sqrt{x^2+\frac{1}{x^2}}} = \int \frac{\frac{(x^2-1)dx}{x^2}}{(x+\frac{1}{x})\sqrt{x^2+\frac{1}{x^2}}}$$

$$\text{SEA } u = x + \frac{1}{x} \rightarrow du = \left(\frac{x^2-1}{x^2}\right)dx \rightarrow x^2 + \frac{1}{x^2} = u^2 - 2$$

$$E = \int \frac{du}{u\sqrt{u^2-2}}, \text{ sea } u = \sqrt{2} \sec \theta$$
$$du = \sqrt{2} \sec \theta \tan \theta d\theta$$



$$E = \int \frac{\sqrt{2} \sec \theta \tan \theta d\theta}{\sqrt{2} \sec \theta \sqrt{2} \tan \theta} = \int d\theta = \theta = \arccos\left(\frac{\sqrt{2}}{u}\right)$$

$$E = \arccos\left(\frac{\sqrt{2}}{x+\frac{1}{x}}\right) = \arccos\left(\frac{x\sqrt{2}}{x^2+1}\right) + C$$

2) CALCULAR

$$E = \int \frac{\sqrt{x} \arcsin(\sqrt{x})}{\sqrt{x^2-x^3}} dx$$

SOLUCION

TRIGONOMETRIC INTEGRALS

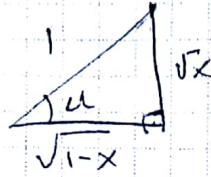
$$E = \int \frac{\sqrt{x} \arcsin(\sqrt{x})}{x \sqrt{1-x}} dx = \int \frac{du \arcsin u}{u^2 \sqrt{1-u^2}}$$

sea $u = \arcsin(\sqrt{x})$

$\sin u = \sqrt{x}$

$\sin^2 u = x$

$2 \sin u \cos u du = dx$



$$E = \int \frac{u \cdot 2 \sin u \cos u}{\sin u \cos u} du = 2 \int u = u^2 = (\arcsin(\sqrt{x}))^2 + C$$

3) CALCULAR

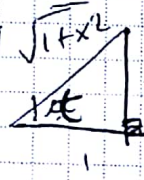
$$E = \int \frac{x e^{\arctan(x)}}{(1+x^2)^{3/2}} dx$$

SOLUCION

SEA $t = \arctan x$

$\tan t = x$

$\sec^2 t dt = dx$



sea $t = \arctan x$

$$E = \int \frac{\tan t e^t \sec^2 t dt}{\sec^3 t} = \int \sin t e^t dt$$

$u = e^t \rightarrow du = e^t dt$

$dv = \sin t dt \rightarrow v = -\cos t$

$$E = -e^t \cos t - \int -\cos t e^t dt$$

$$E = -e^t \cos t + \int \cos t e^t dt$$

$u = e^t \rightarrow du = e^t dt$

$dv = \cos t dt \rightarrow v = \sin t$

$$E = -e^t \cos t + [e^t \sin t - \int \sin t e^t dt]$$

$$E = -e^t \cos t + e^t \sin t - E$$

$$2E = e^t (\sin t - \cos t)$$

$$E = \frac{e^t}{2} (\sin t - \cos t) = \frac{e^{\arctan x}}{2} (\sin(\arctan x) - \cos(\arctan x)) + C$$

$$E = \int \frac{[x + x \tan^2(x)] dx}{3 + 4 \tan(x) + \sec^2(x)}$$

SOLUCION

TRANSFORMANDO EL INTEGRANDO

$$E = \int \frac{x(1 + \tan^2(x)) dx}{\tan^2(x) + 4 \tan x + 4} = \int \frac{x \sec^2 x dx}{(\tan x + 2)^2}$$

$$u = x \rightarrow du = dx$$

$$dv = \frac{\sec^2 x dx}{(\tan x + 2)^2} \rightarrow v = \int \frac{d\lambda}{(\lambda + 2)^2} = -\frac{1}{\lambda + 2} = -\frac{1}{\tan x + 2}$$

$$E = -\frac{x}{\tan x + 2} - \int -\frac{1}{\tan x + 2} dx = -\frac{x}{\tan x + 2} + \int \frac{dx}{\tan x + 2}$$

$$\text{Como } R(\sec x, -\cos x) = R(\sec x, \cos x)$$

$$\text{HACEMOS } t = \tan x \rightarrow dx = \frac{dt}{1+t^2}$$

$$E = -\frac{x}{\tan x + 2} + \int \frac{\frac{dt}{1+t^2}}{t+2} = -\frac{x}{\tan x + 2} + \int \frac{dt}{(t+2)(t^2+1)}$$

$$*) \frac{1}{(t+2)(t^2+1)} = \frac{A}{t+2} + \frac{Bt+C}{t^2+1}, \quad A = \frac{1}{5}$$

$$1 = A(t+1) + (Bt+C)(t+2)$$

$$\text{para } t=i \rightarrow 1 = (Bi+C)(i+2) = \underbrace{(2B+C)i}_0 + \underbrace{(2C-B)}_1$$

$$\begin{cases} 2B+C=0 \\ 2C-B=1 \end{cases} \rightarrow B = -\frac{4}{5}, \quad C = \frac{2}{5}$$

$$E = -\frac{x}{\tan x + 2} + \frac{1}{5} \int \frac{dt}{t+2} + \int \frac{-\frac{4}{5} + \frac{2}{5}}{t^2+1} dt$$

$$E = -\frac{x}{\tan x + 2} + \frac{1}{5} \ln|t+2| - \frac{1}{5} \left[\int \frac{t dt}{t^2+1} - 2 \int \frac{dt}{t^2+1} \right]$$

$$E = -\frac{x}{\tan x + 2} + \frac{1}{5} \ln|\tan x + 2| - \frac{1}{10} \ln|t^2+1| + \frac{2}{5} \arctan(t)$$

$$E = -\frac{x}{\tan x + 2} + \frac{1}{5} \ln|\tan x + 2| - \frac{1}{10} \ln(\tan^2 x + 1) + \frac{2}{5} \arctan(\tan x)$$

$$E = -\frac{x}{\tan x + 2} + \frac{1}{5} \ln|\tan x + 2| - \frac{1}{10} \ln(\tan^2 x + 1) + \frac{2}{5} x + C$$

Q. Evaluate

$$E = \int \frac{[\tan(x) + \tan(x)] dx}{[4 + \sec^2(x)]^3 [\tan^2 x + 8]^4}$$

SOLUTION

$$E = \int \frac{\tan x (1 + \tan^2 x) dx}{[4 + 1 + \tan^2 x]^3 [\tan^2 x + 8]^4}$$

$$\text{Let } u = \tan^2 x$$

$$du = 2 \tan x \sec^2 x dx$$

$$E = \frac{1}{2} \int \frac{2 \tan x \sec^2 x dx}{[\tan^2 x + 5]^3 [\tan^2 x + 8]^4}$$

$$E = \frac{1}{2} \int \frac{du}{(u+5)^3 (u+8)^4}$$

$$z = \frac{u+5}{u+8} \rightarrow u = \frac{5-8z}{z-1} \rightarrow du = \frac{3dz}{(z-1)^2}$$

$$\text{*) } u+5 = \frac{-3z}{z-1} \quad \text{*) } u+8 = \frac{-3}{z-1}$$

$$E = \frac{1}{2} \int \frac{\frac{3dz}{(z-1)^2}}{\left(\frac{-3z}{z-1}\right)^3 \left(\frac{-3}{z-1}\right)^4} = -\frac{1}{2(3^6)} \int \frac{(z-1)^5 dz}{z^3}$$

$$= -\frac{1}{2(3^6)} \int \frac{z^5 - 5z^4 + 10z^3 - 10z^2 + 5z - 1}{z^3} dz$$

$$= -\frac{1}{2(3^6)} \int \left[z^2 - 5z + 10 - \frac{10}{z} + \frac{5}{z^2} - \frac{1}{z^3} \right] dz$$

$$= -\frac{1}{2(3^6)} \left[\frac{z^3}{3} - \frac{5}{2} z^2 + 10z - 10 \ln z - \frac{5}{z} + \frac{1}{2z^2} \right]$$

$$= -\frac{1}{2(3^6)} \left[\frac{1}{3} \left(\frac{u+5}{u+8} \right)^3 - \frac{5}{2} \left(\frac{u+5}{u+8} \right)^2 + 10 \left(\frac{u+5}{u+8} \right) - 10 \ln \left| \frac{u+5}{u+8} \right| - 5 \left(\frac{u+8}{u+5} \right) + \frac{1}{2} \left(\frac{u+8}{u+5} \right)^2 \right]$$

WHERE $u = \tan^2 x$



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6) CALCULAR

$$E = \int \frac{(3e^{12x} - 2e^{6x}) dx}{e^{12x} - 5e^{6x} + 6}$$

SOLUCION
sea $u = e^{6x} \rightarrow du = e^{6x} 6 dx = 6u dx \rightarrow dx = \frac{du}{6u}$

$$E = \int \frac{(3u^2 - 2u) \frac{du}{6u}}{u^2 - 5u + 6} = \int \frac{u(3u-2) \frac{du}{6u}}{(u-3)(u-2)}$$

$$E = \frac{1}{6} \int \frac{(3u-2) du}{(u-3)(u-2)}$$

$$1) \frac{3u-2}{(u-3)(u-2)} = \frac{A}{u-3} + \frac{B}{u-2}, \quad A=7, \quad B=-4$$

$$E = \frac{1}{6} \left[7 \int \frac{du}{u-3} - 4 \int \frac{du}{u-2} \right] = \frac{1}{6} [7 \ln|u-3| - 4 \ln|u-2|]$$

$$E = \frac{7}{6} \ln|e^{6x}-3| - \frac{4}{6} \ln|e^{6x}-2| + C$$

7) CALCULAR

$$E = \int \frac{[\ln^3(x) + 2\ln x - 1] dx}{x [\ln^2 x - 4\ln x + 8]}$$

SOLUCION

~~E =~~ $u = \ln x \rightarrow du = \frac{dx}{x}$

OTRO PROBLEMA

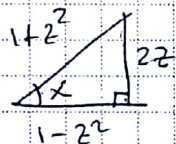
$$\begin{aligned}
 & \int \frac{1}{u^2 - 4u + 8} du \\
 E &= \int (u+4) du + \int \frac{10u-33}{u^2-4u+8} du = \frac{u^2}{3} + 4u + \int \frac{5(2u-4) - 13}{u^2-4u+8} du \\
 &= \frac{u^2}{3} + 4u + 5 \int \frac{(2u-4) du}{u^2-4u+8} - 13 \int \frac{du}{u^2-4u+8} \\
 &= \frac{u^2}{3} + 4u + 5 \ln |u^2-4u+8| - 13 \int \frac{du}{(u-2)^2 + 2^2} \\
 &= \frac{u^2}{3} + 4u + 5 \ln |u^2-4u+8| - \frac{13}{2} \int \frac{d(\frac{u-2}{2})}{(\frac{u-2}{2})^2 + 1} \\
 &= \frac{u^2}{3} + 4u + 5 \ln |u^2-4u+8| - \frac{13}{2} \arctan \left(\frac{u-2}{2} \right) \\
 &= \frac{(\ln x)^2}{3} + 4 \ln x + 5 \ln |(\ln x)^2 - 4 \ln x + 8| - \frac{13}{2} \arctan \left(\frac{\ln x - 2}{2} \right) + C
 \end{aligned}$$

8) CALCULAR

$$E = \int \frac{\cos x \, dx}{3 - 3 \sin x + 2 \cos x}$$

SOLUCION

$$z = \tan\left(\frac{x}{2}\right) \rightarrow dx = \frac{2dz}{1+z^2}$$



$$\sin x = \frac{z}{\sqrt{1+z^2}}, \quad \cos x = \frac{1}{\sqrt{1+z^2}}$$

$$E = \int \frac{\frac{1-z^2}{1+z^2} \cdot \frac{2dz}{1+z^2}}{3 - 3\left(\frac{z}{\sqrt{1+z^2}}\right) + 2\left(\frac{1}{\sqrt{1+z^2}}\right)} = \int \frac{2(1-z^2) dz}{(1+z^2)^2 (3 + 3z^2 - 6z + 2 - 2z^2)}$$

$$E = 2 \int \frac{(1-z^2) dz}{(z^2 - 6z + 5)(1+z^2)} = 2 \int \frac{(1-z^2) dz}{(z-5)(z-1)(z^2+1)}$$

$$E = -2 \int \frac{(z+1)(z-1) dz}{(z-5)(z-1)(z^2+1)} = -2 \int \frac{(z+1) dz}{(z-5)(z^2+1)}$$

$$x) \frac{z+1}{(z-5)(z^2+1)} = \frac{A}{z-5} + \frac{Bz+C}{z^2+1} \rightarrow A = \frac{6}{26} = \frac{3}{13}$$

$$\begin{aligned}
 x) z+1 &= A(z^2+1) + (z-5)(Bz+C) \\
 z+1 &= A(0) + (z-5)(Bz+C) \\
 z+1 &= -B + zC - 5Bz - 5C
 \end{aligned}
 \quad \left| \begin{aligned}
 z+1 &= (C-5B)z + (-B-5C) \\
 \left. \begin{aligned} C-5B &= 1 \\ -B-5C &= 1 \end{aligned} \right\} \begin{aligned} 5C-25B &= 1 \\ -B-5C &= 1 \end{aligned} \end{aligned} \right.$$

$$E = -2 \left[A \int \frac{dz}{z-s} + \int \frac{Bz+C}{z^2+1} dz \right] = -2 \left[\frac{3}{13} \ln|z-s| + \int \frac{\frac{3}{13}z - \frac{2}{13}}{z^2+1} dz \right]$$

$$E = -2 \left[\frac{3}{13} \ln|z-s| - \frac{1}{13} \int \frac{3z+2}{z^2+1} dz \right]$$

$$E = -2 \left[\frac{3}{13} \ln|z-s| - \frac{1}{13} \left[3 \int \frac{z dz}{z^2+1} + 2 \int \frac{dz}{z^2+1} \right] \right]$$

$$E = -\frac{6}{13} \ln|z-s| + \frac{2}{13} \left[\frac{3}{2} \int \frac{2z dz}{z^2+1} + 2 \arctan(z) \right]$$

$$E = -\frac{6}{13} \ln|z-s| + \frac{3}{13} \ln(z^2+1) + \frac{4}{13} \arctan(z)$$

$$E = -\frac{6}{13} \ln \left| \tan\left(\frac{x}{2}\right) - s \right| + \frac{3}{13} \ln(\tan^2(x/2) + 1) + \frac{4}{13} \arctan(\tan(x/2))$$

$$E = -\frac{6}{13} \ln \left| \tan\left(\frac{x}{2}\right) - s \right| + \frac{3}{13} \ln(\tan^2(x/2) + 1) + \frac{2x}{13} + C$$

7) CALCULAR

$$E = \int \frac{[\ln^3(x) + 2\ln(x) - 1] dx}{x [\ln^2 x - 4\ln x + 8]^4}$$

SOLUCION

$$u = \ln x \rightarrow du = \frac{dx}{x}$$

$$E = \int \frac{(u^3 + 2u - 1) du}{(u^2 - 4u + 8)^4} = \int \frac{u^3 + 2u - 1}{[(u-2)^2 + 4]^4} du$$

$$ax^2 + bx + c = a[(x-p)^2 + q^2] \rightarrow x-p = q\tau, \quad dx = q d\tau$$

$$u-2 = 2\tau \rightarrow du = 2d\tau$$

$$u = 2\tau + 2$$

$$E = \int \frac{(2\tau+2)^3 + 2(2\tau+2) - 1}{[(2\tau)^2 + 4]^4} \cdot 2d\tau$$

$$F_3 = \frac{1}{(t^2+1)^3}$$

$$= \underbrace{\frac{1}{32} \int \frac{8t^3 + 28t}{(t^2+1)^3} dt}_{F_1} + \underbrace{\frac{1}{32} \int \frac{24t^2 + 11}{(t^2+1)^3} dt}_{F_2}$$

> 1) CALCULO DE F_1 ($m=3$)

$$t^2+1 = u \rightarrow 2t dt = du \rightarrow t^2 = u-1$$

$$F_1 = \frac{1}{32} \int \frac{8t^3 + 28t}{(t^2+1)^3} dt = \frac{8}{32} \int \frac{t^3 dt}{(t^2+1)^3} + \frac{28}{32} \int \frac{t dt}{(t^2+1)^3}$$

$$= \frac{1}{4} \int \frac{t^3 (2t dt)}{(t^2+1)^3} + \frac{7}{8} \int \frac{2t dt}{(t^2+1)^3}$$

$$= \frac{1}{8} \int \frac{(u-1) du}{u^3} + \frac{7}{16} \int \frac{du}{u^2} = \frac{1}{8} \int \left(\frac{1}{u^2} - \frac{1}{u^3} \right) du + \frac{7}{16} \int \frac{1}{u^2} du$$

$$= -\frac{1}{8} \frac{1}{u} - \frac{1}{8} \left(-\frac{1}{2} \frac{1}{u^2} \right) + \frac{7}{16} \left(-\frac{1}{2} \frac{1}{u^2} \right)$$

$$= -\frac{1}{8} \cdot \frac{1}{u} + \frac{1}{16} \cdot \frac{1}{u^2} - \frac{7}{32} \cdot \frac{1}{u^2} = -\frac{1}{8u} - \frac{5}{32u^2} = -\frac{1}{8} \frac{1}{(t^2+1)} - \frac{5}{32} \frac{1}{(t^2+1)^2}$$

$$= -\frac{1}{8} \frac{1}{\frac{(x-2)^2}{4} + 1} - \frac{5}{32} \frac{1}{\left[\frac{(x-2)^2}{4} + 1 \right]^2} \quad dt = \sec^2 u du$$

2) CALCULO DE F_2 ($m=2$), $t = \tan u \rightarrow dt = \sec^2 u du$ - $u = \arctan t$

$$F_2 = \frac{1}{32} \int \frac{24t^2 + 11}{(t^2+1)^3} dt = \frac{1}{32} \int \frac{24(t^2+1) - 13}{(t^2+1)^3} dt$$

$$= \frac{24}{32} \int \frac{dt}{(t^2+1)^2} - \frac{13}{32} \int \frac{dt}{(t^2+1)^3} = \frac{3}{4} \int \frac{\sec^2 u du}{\sec^4 u} - \frac{13}{32} \int \frac{\sec^2 u du}{\sec^6 u}$$

$$= \frac{3}{4} \int \cos^2 u du - \frac{13}{32} \int \cos^4 u du = \frac{3}{4} \int \left(\frac{1 + \cos 2u}{2} \right) du$$

$$- \frac{13}{32} \int \left(\frac{1 + \cos 2u}{2} \right)^2 du = \frac{12}{32} \left[\int du + \int \cos 2u du \right] - \frac{13}{32(4)} \int (1 + 2 \cos 2u + \cos^2 2u) du$$